Dataset Overview

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2024 analytical dataset

Field	Value
Reporting year	2024
Geographic level	Bremen Ortsteile
Total Ortsteile	88
Included	79
Excluded	9
Baseline indicators	3

Source counts; source-level missing observations

Source	Records	Missing	Reporting years
excel_ortsteilatlas_long.csv	316064	24355.000	1969-2025; indicator-specific
csv_district_statistics_long.csv	15147	223.000	2010-2016
bremen_data.json	1515	—	Catalogue metadata

2024 baseline coverage out of 88 Ortsteile

Indicator	Observed	Missing
Unemployment	79	9
Employment	79	9
SGB II under 65	84	4

Complete-case scoring; no imputation. Rates and charts use the same 79 Ortsteile unless labeled otherwise.

Indicators and Descriptive Statistics

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Baseline indicators; source: Datentabelle.xls

Original German definition	English name	Unit	Year	Direction
Anteil der Arbeitslosen an den sozialversicherungspflichtig Beschäftigten zuzüglich Arbeitslose (%)	Unemployment quotient (%)	%	2024	higher
Anteil der sozialversicherungspflichtig Beschäftigten an der Bevölkerung im Alter von 15 bis unter 65 Jahren (%)	Social-insurance employment (%)	%	2024	lower
Anteil der Leistungsberechtigten (Grundsicherung für Arbeitsuchende, SGB II) an der Bevölkerung unter 65 Jahren (%)	SGB II receipt, under 65 (%)	%	2024	higher

Unemployment denominator: social-insurance employees + registered unemployed. Employment denominator: population aged 15 to under 65. SGB II denominator: population under 65.

2024; percent; unweighted Ortsteile; SD uses ddof=1

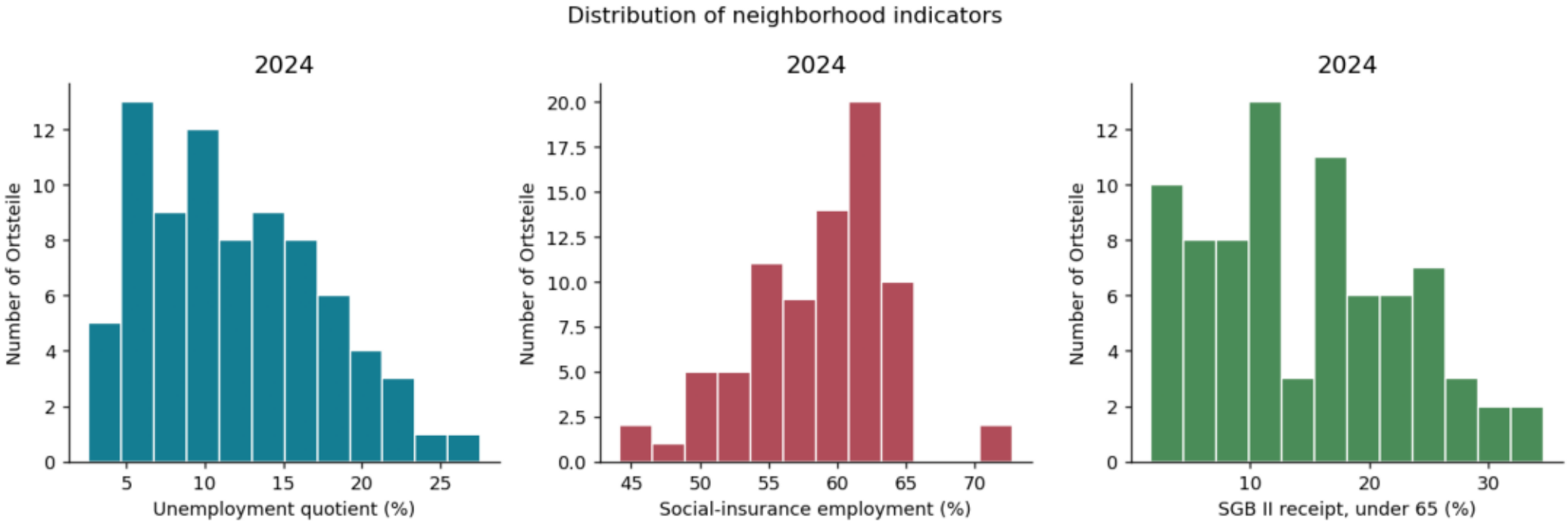
Indicator	n	Mean	Median	SD	Minimum	Maximum
Unemployment	79.000	11.862	10.976	5.554	2.590	27.541
Employment	79.000	58.630	59.404	5.313	44.159	72.714
SGB II under 65	79.000	14.844	13.232	8.392	1.702	34.622

Spearman indicator correlations; unitless

Indicator	Unemployment	Employment	SGB II under 65
Unemployment	1.000	-0.667	0.974
Employment	-0.667	1.000	-0.588
SGB II under 65	0.974	-0.588	1.000

Indicator Distributions

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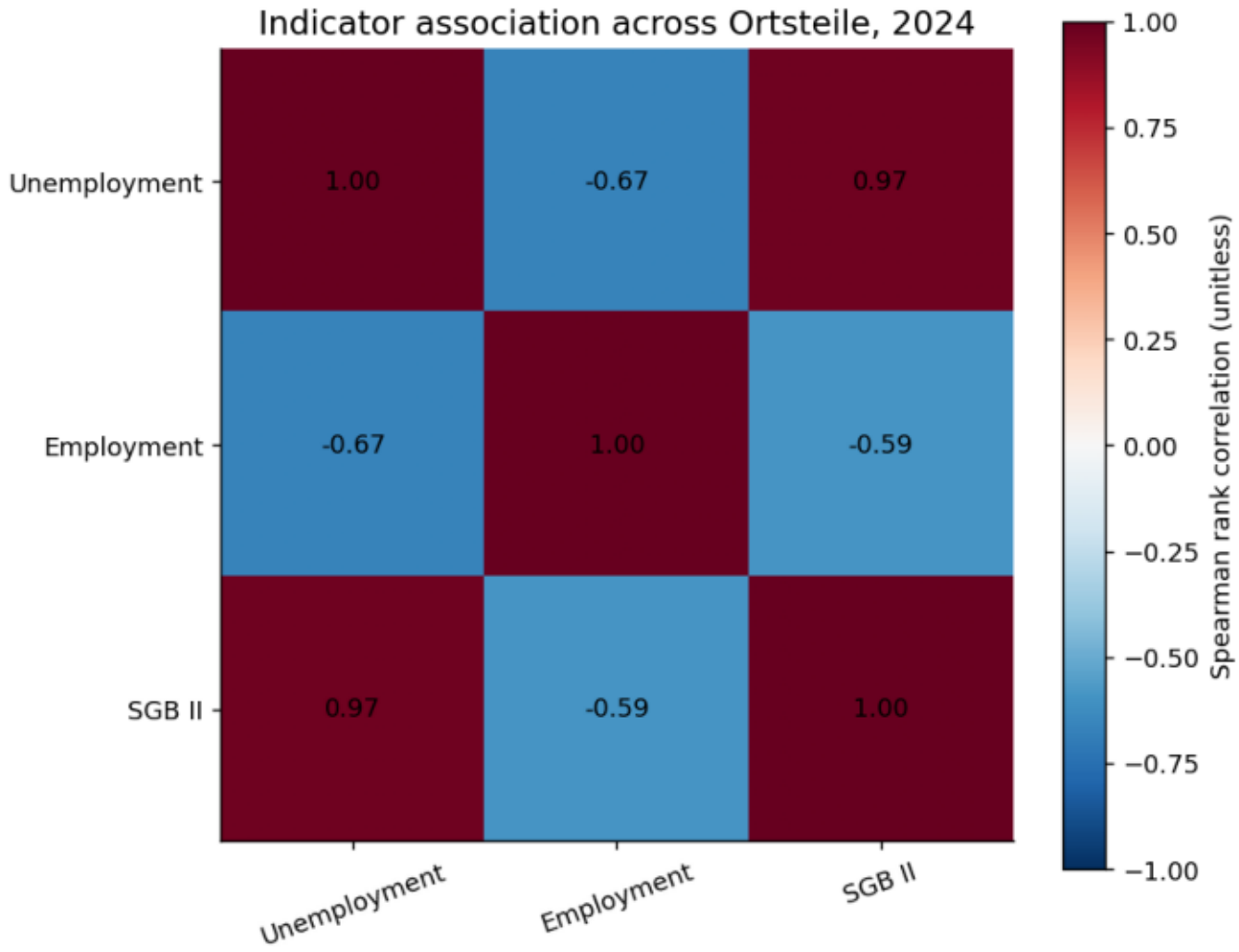


Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Complete cases; unweighted neighborhoods; descriptive, not causal.

2024; percent; n=79.

Indicator Correlations

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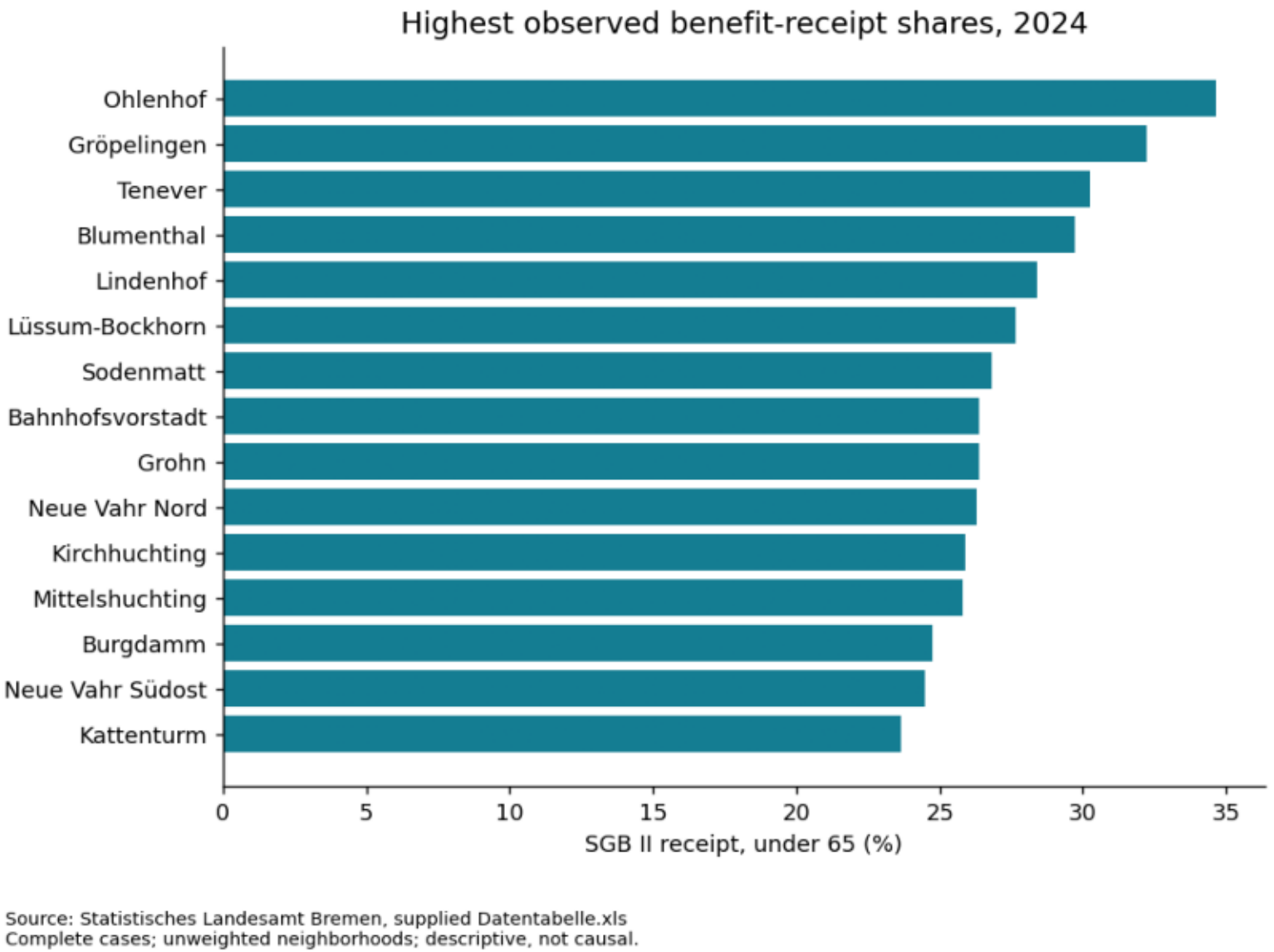


Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Correlation does not imply causation; indicators overlap conceptually.

2024 Spearman correlation; unitless; n=79.

Neighborhood Comparison

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2024 under-65 SGB II receipt; percent; highest 15 observed shares.

Composite Index: Selected Rankings

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Top 10; 2024; score 0-1; rank 1 = greatest measured disadvantage

Rank	Code	Ortsteil	Score
1	443	Ohlenhof	0.943
2	442	Gröpelingen	0.876
3	522	Grohn	0.821
4	531	Blumenthal	0.795
5	441	Lindenhof	0.777
6	373	Tenever	0.769
7	533	Lüssum-Bockhorn	0.724
8	243	Kirchhuchting	0.708
9	112	Bahnhofsvorstadt	0.698
10	525	Fähr-Lobbendorf	0.693

Bottom 5; 2024; descending ranks retained

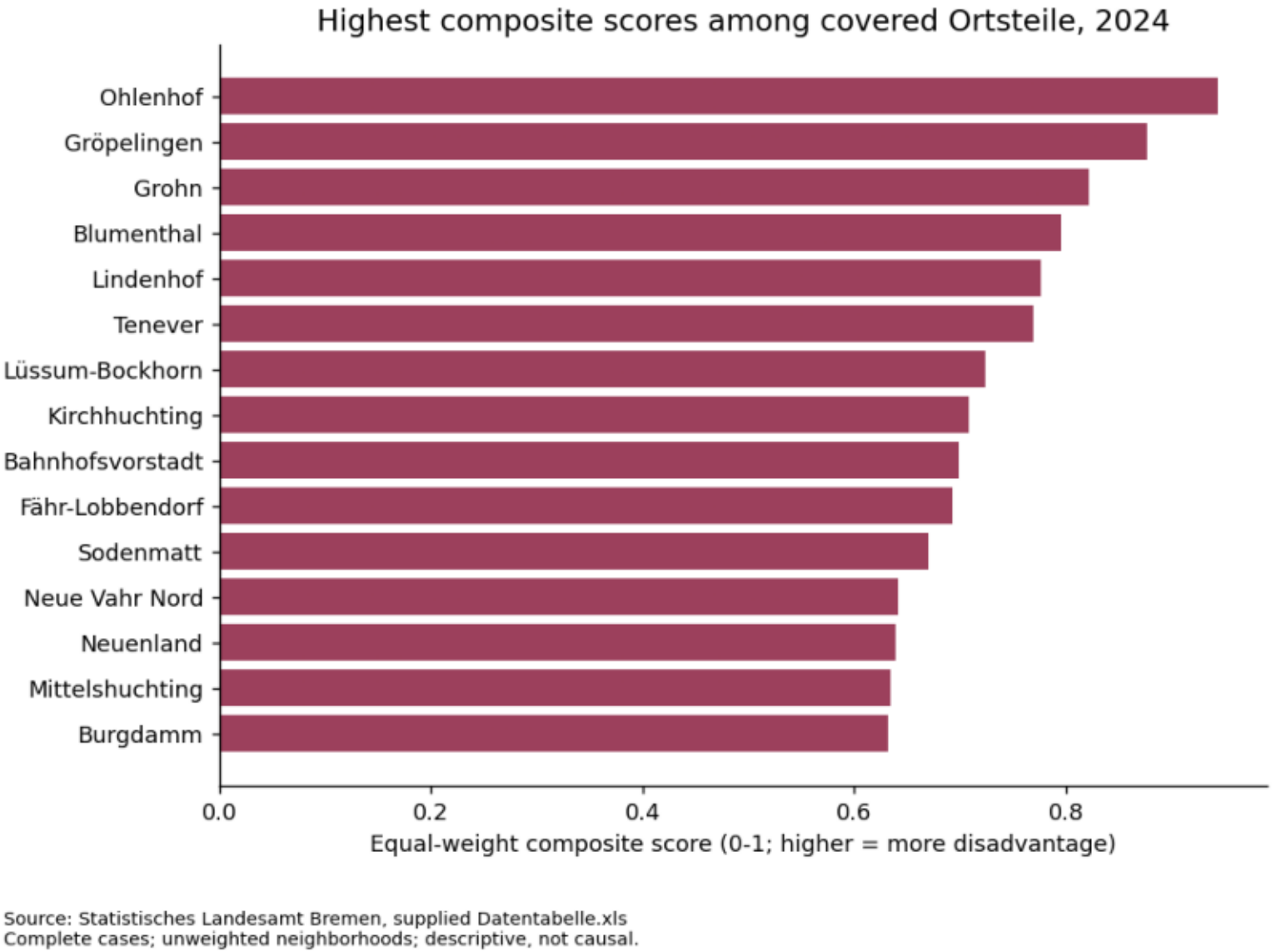
Rank	Code	Ortsteil	Score
75	323	Barkhof	0.170
76	244	Grolland	0.165
77	374	Osterholz	0.137
78	231	Habenhausen	0.136
79	261	Seehausen	0.012

Normalization: (x-min)/(max-min); employment reversed. Weights: 1/3 per indicator. Descending competition ranks.

Source: supplied municipal datasets; saved verified outputs.

Composite Index: Ranking Chart

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2024 equal-weight composite scores; highest 15; score 0-1.

TOPSIS: Selected Rankings

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Top 10; 2024; disadvantage closeness 0-1

Rank	Code	Ortsteil	Score
1	443	Ohlenhof	0.971
2	442	Gröpelingen	0.911
3	531	Blumenthal	0.810
4	441	Lindenhof	0.796
5	373	Tenever	0.767
6	522	Grohn	0.738
7	533	Lüssum-Bockhorn	0.726
8	112	Bahnhofsvorstadt	0.726
9	217	Neuenland	0.699
10	243	Kirchhuchting	0.696

Bottom 5; 2024; descending ranks retained

Rank	Code	Ortsteil	Score
75	326	Schwachhausen	0.093
76	351	Borgfeld	0.093
77	374	Osterholz	0.085
78	231	Habenhausen	0.070
79	261	Seehausen	0.024

Vector norm; equal 1/3 weights; Euclidean distances; ideal: high unemployment/SGB II, low employment; C=D-/(D+ + D-).

Source: supplied municipal datasets; saved verified outputs.

K-means: Profile Statistics

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Silhouette diagnostics; selected k=2

k	silhouette	minimum cluster size
2	0.528	28
3	0.384	20
4	0.347	13
5	0.331	9
6	0.355	4
7	0.367	4
8	0.389	2

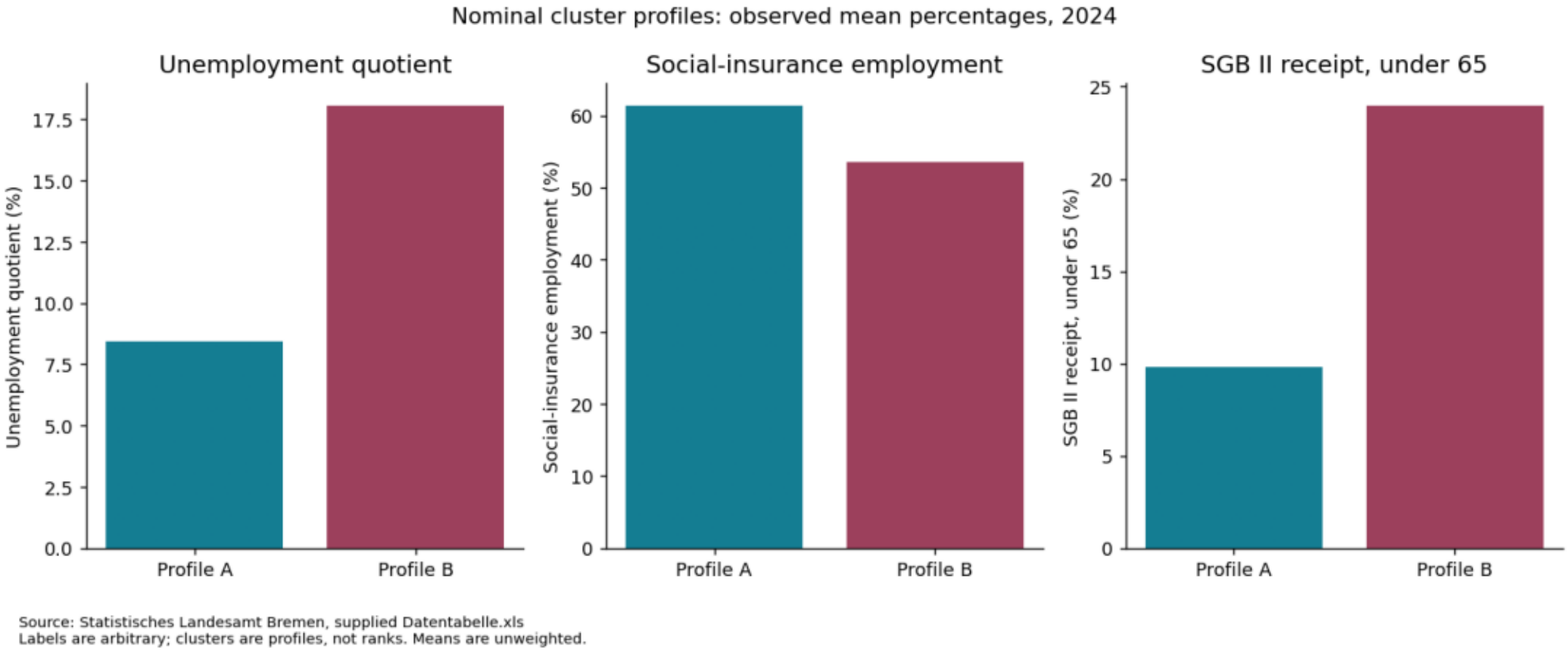
2024 profile sizes and mean indicator percentages

Profile	n	Unemployment (%)	Employment (%)	SGB II under 65 (%)
Profile A	51	8.449	61.404	9.823
Profile B	28	18.077	53.579	23.989

Nominal labels. StandardScaler (ddof=0); k-means++; n_init=30; seed=42. Tested k=2..8; maximum silhouette subject to minimum profile size 3.

K-means: Profile Chart

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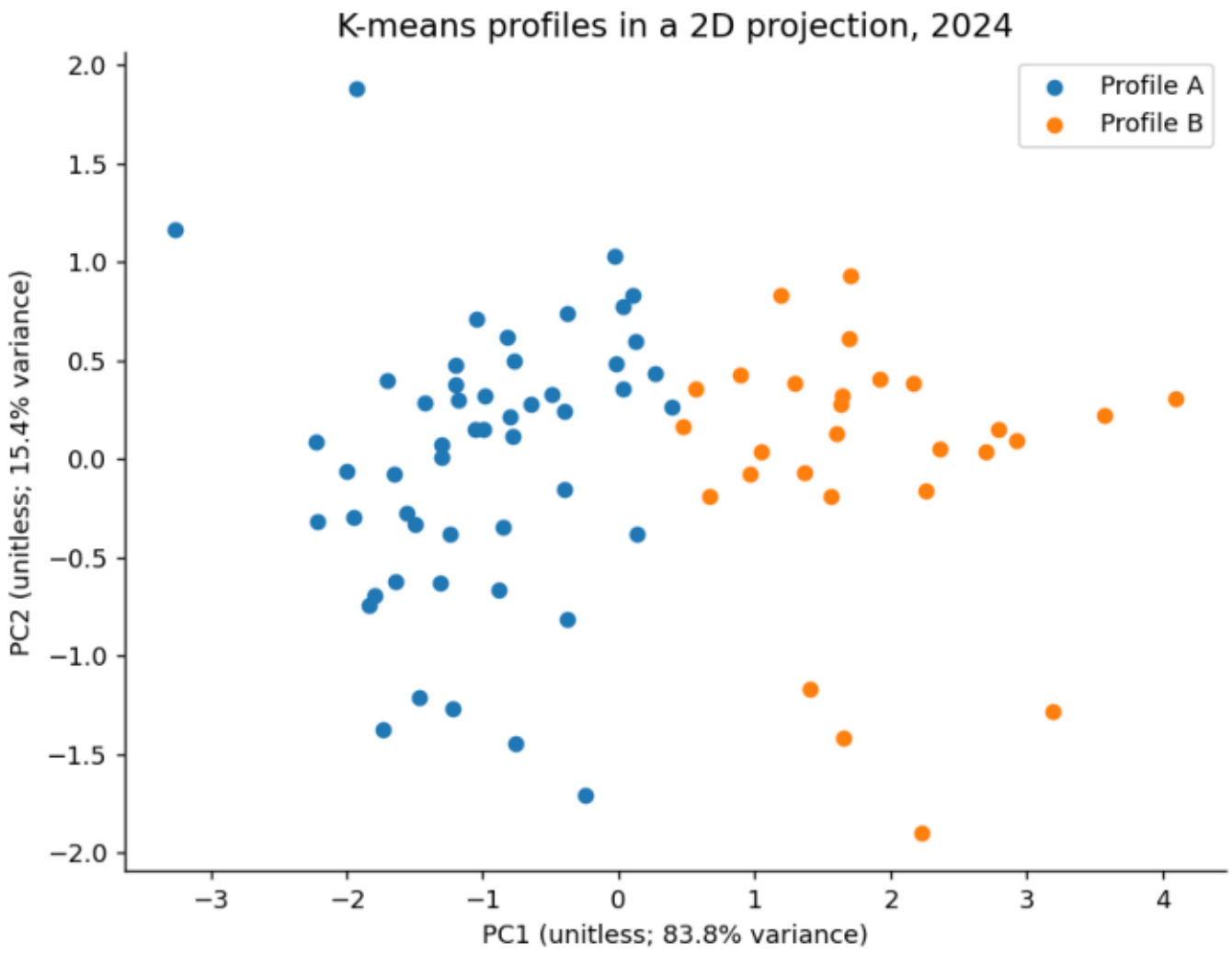


Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Labels are arbitrary; clusters are profiles, not ranks. Means are unweighted.

2024 profile means; percent; labels are nominal.

K-means: PCA Projection

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2024; standardized coordinates; PCA display only. K-means inputs: three standardized indicators.

Ranking and Method Comparison

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2024 rank correlations

Spearman	Kendall tau-b	n
0.974	0.883	79

Five largest absolute rank disagreements

Ortsteil	Composite rank	TOPSIS rank	TOPSIS - composite	Absolute difference
Rablinghausen	73.000	50.000	-23.000	23.000
St. Magnus	38.000	57.000	19.000	19.000
Lehe	49.000	63.000	14.000	14.000
Fähr-Lobbendorf	10.000	20.000	10.000	10.000
Oberneuland	60.000	69.000	9.000	9.000

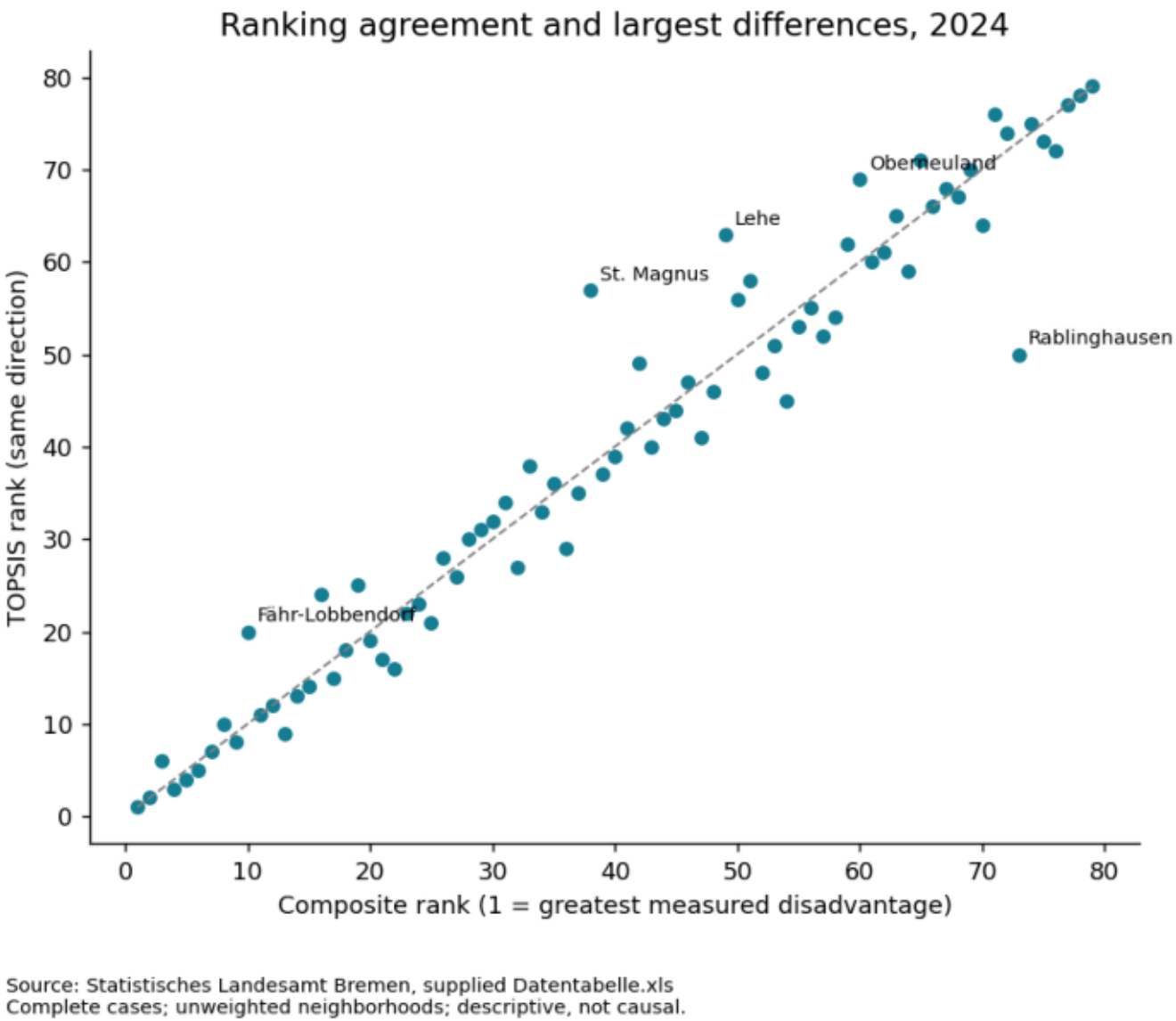
Method outputs and settings

Method	Output	Settings
Composite	Score [0,1]; descending rank	Directed min-max; equal 1/3 weights
TOPSIS	Closeness [0,1]; descending rank	Vector norm; equal 1/3 weights; Euclidean distance
K-means	2 nominal profiles	StandardScaler; n_init=30; random_state=42

Source: supplied municipal datasets; saved verified outputs.

Ranking Comparison: Scatter Plot

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2024; n=79; rank 1 = greatest measured disadvantage.

Sensitivity: Alternative Weights

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Weight vectors: unemployment / employment / SGB II

Scenario	Unemployment	Employment	SGB II under 65
Equal weights	0.333	0.333	0.333
Domain-balanced	0.250	0.250	0.500
Unemployment emphasis	0.500	0.250	0.250
Employment emphasis	0.250	0.500	0.250

Baseline and alternative-weight statistics; n=79; same row set

Scenario	Method	Spearman	Mean abs. movement	Max movement	Top-10 overlap
Equal weights	composite	1.000	0.000	0.000	10
Equal weights	topsis	1.000	0.000	0.000	10
Domain-balanced	composite	0.994	1.722	8.000	9
Domain-balanced	topsis	0.995	1.468	10.000	9
Unemployment emphasis	composite	0.997	1.190	7.000	10
Unemployment emphasis	topsis	0.997	1.241	5.000	10
Employment emphasis	composite	0.986	2.759	11.000	9
Employment emphasis	topsis	0.996	1.266	8.000	9

Source: supplied municipal datasets; saved verified outputs.

Sensitivity: Feature Sets and Cluster Stability

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Feature-selection scenarios; n=79

Feature scenario	Method	Spearman	Max movement	Top-10 overlap
omit_unemployment	composite	0.991	9.000	9
omit_unemployment	topsis	0.993	10.000	9
omit_employment	composite	0.968	24.000	8
omit_employment	topsis	0.999	4.000	10
omit_benefit_receipt	composite	0.972	13.000	10
omit_benefit_receipt	topsis	0.992	12.000	10
include_child_benefits	composite	0.993	7.000	9
include_child_benefits	topsis	0.994	7.000	9

200 constrained random weight draws per method; seed=2024

Method	Draws	Minimum Spearman	Max movement	Minimum top-10 overlap
composite	200	0.962	18.000	9
topsis	200	0.981	18.000	8

K-means refit stability: adjusted Rand index

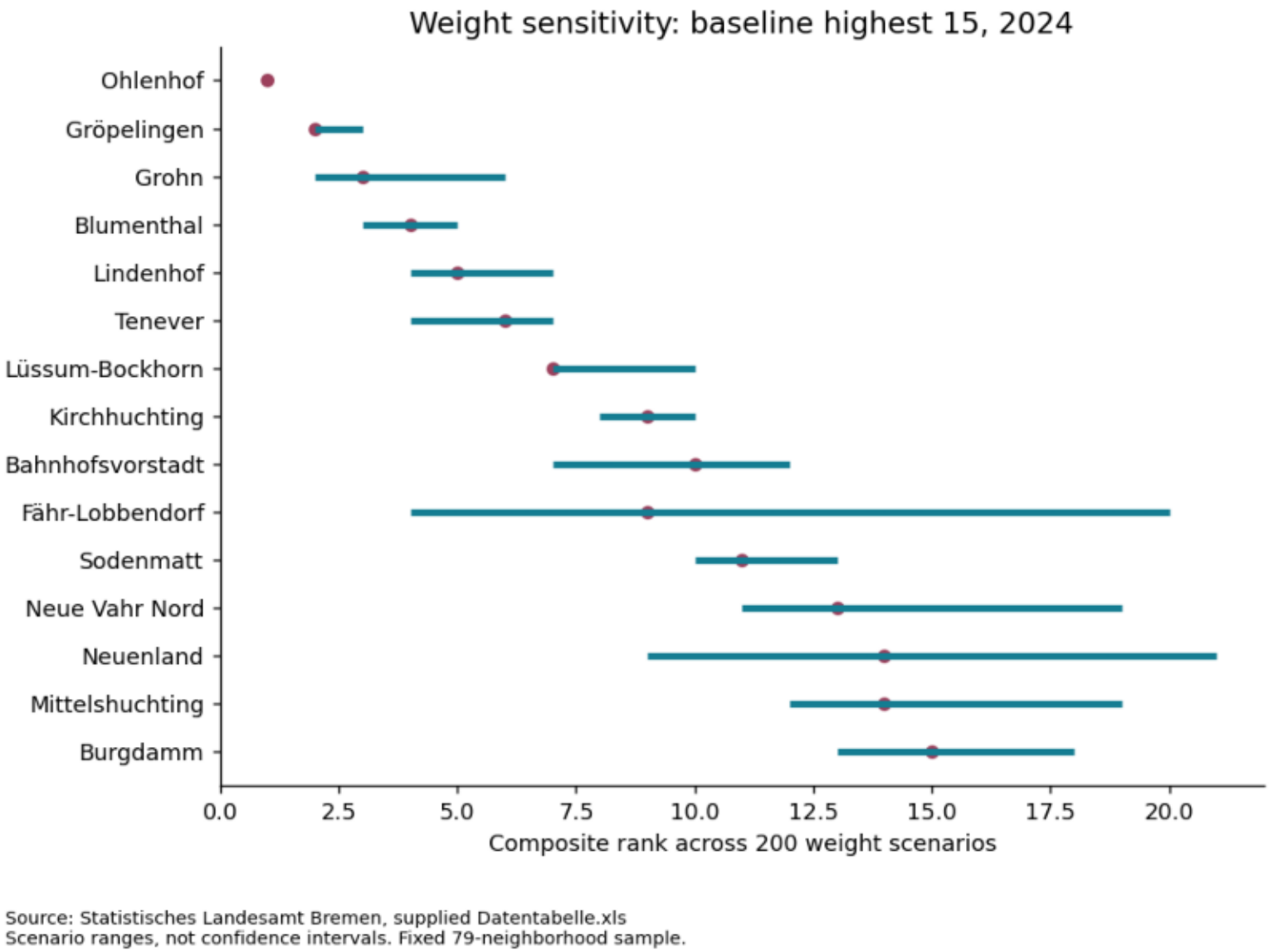
Refit	Mean ARI	Min ARI	Max ARI
bootstrap_neighborhoods	0.892	0.591	1.000
initialization	1.000	1.000	1.000

Clustering: 30 initialization seeds and 30 bootstrap refits. Random weights: Dirichlet(3,3,3), each weight in [0.10,0.60].

Source: supplied municipal datasets; saved verified outputs.

Sensitivity: Rank-Range Visualization

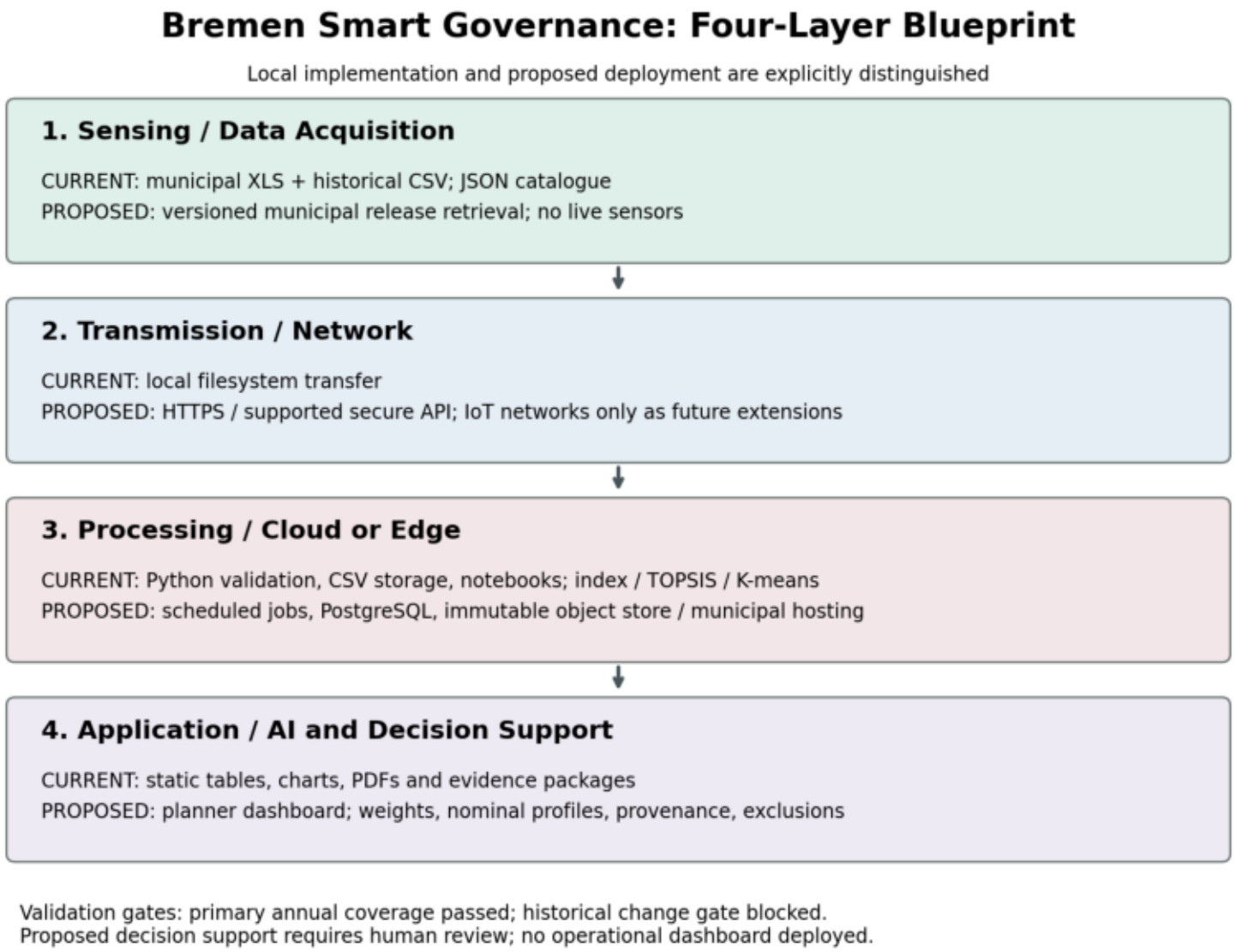
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2024; 200 weight scenarios; baseline highest 15; scenario ranges, not confidence intervals.

Four-Layer Architecture

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Existing four-layer diagram; implemented and proposed components labeled.

Source: supplied municipal datasets; saved verified outputs.

Sources and Reproducibility

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Dataset references

Source	File / dataset	Reporting period
Primary statistics	Datentabelle.xls; Bremer Ortsteilatlas	2024 baseline
Historical statistics	22 ST*.csv files	2010-2016
Discovery metadata	bremen_data.json	Not observations

Complete outputs and reusable scripts

Artifact	Repository-relative path
Complete notebook/PDF	Bremen_Smart_Governance_COMPLETE_ANALYSIS.ipynb / .pdf
Full scores, rankings, profiles	results/composite_index_results.csv; topsis_results.csv; kmeans_results.csv
Full sensitivity rows	results/sensitivity_results.csv
Preparation / validation	src/prepare_data.py; src/build_analysis_dataset.py; src/verify_outputs.py
Statistics / scoring	src/descriptive_analysis.py; src/composite_index.py; src/topsis.py
Clustering / sensitivity	src/clustering.py; src/sensitivity_analysis.py
Settings / dependencies	docs/methods.md; requirements.txt; docs/environment_versions.json
Regenerate this summary	python src/create_results_summary.py

Official atlas reference: https://www.statistik.bremen.de/datenangebote/bremer-ortsteilatlas-15228

Historical change estimates not included: boundary and definition comparability checks remain incomplete. Exact within-year reference dates remain unverified. Display: 3 decimals; stored result precision unchanged.